

NEURAX 2.0 – Abstract Submission

Team Name: Soul Reaper's
Theme name: HealthTech & Med AI
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Project Title:

Khushi Hygieia: An Integrated Digital Healthcare Ecosystem for Inclusive and Regional Medical Access.

Theme Selected:

HealthTech & Med AI

Problem Statement:

Design and develop an AI-powered remote healthcare platform that enables early diagnosis, predictive risk monitoring, secure patient data handling, and automated hospital workflow optimization — specifically targeting rural and underserved populations.

Proposed Solution:

Khushi Hygieia is a centralized digital healthcare platform designed to bridge the gap between regional communities and advanced medical infrastructure. The system integrates multiple healthcare services into a single interface including a Smart Hospital Finder that uses GPS to locate nearby hospitals, a Blood Donation Network that connects donors with patients in need, and multilingual medical guidance to assist users in English, Hindi, and Telugu. The platform also includes teleconsultation functionality allowing patients to connect with doctors remotely, reducing travel barriers and enabling faster medical guidance.

Technical Approach:

The platform is developed using a modern full stack architecture. The frontend is built using React.js with Vite for high performance and responsive design. Supabase serves as the backend as a service providing authentication, database management, and Realtime synchronization with role-based security policies. Artificial intelligence capabilities are integrated through the Google Gemini API to power the symptom checker and medical assistant features. Teleconsultation is implemented using PeerJS and WebRTC for secure peer to peer video communication. The interface uses optimized styling and animations to provide an accessible and user-friendly experience across devices.

Expected Impact:

Khushi Hygieia aims to improve healthcare accessibility by connecting patients with nearby hospitals, blood donors, and medical professionals through a unified platform. The system reduces emergency response time, enables remote consultations for rural populations, and provides multilingual medical guidance that improves patient understanding and treatment adherence. With scalable cloud infrastructure, the platform can be expanded to support larger healthcare networks and integrate with public health systems, contributing to a more inclusive and efficient digital healthcare ecosystem.